

Department of Statistics and Data Science
The Chinese University of Hong Kong

Gifted Education Fund: Off-school Advanced Learning Programmes

Programme 2025-04

Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

Phase 4 Capstone Project Development

From Risk Learning to Showcase Presentation

12 August 2026, Wednesday, 13:30-16:00

L1, Institute of Chinese Studies, The Chinese University of Hong Kong

Why are we here today?

This session starts **Phase 4: Capstone Project Development**. In Phases 1 to 3, you learned ideas related to risk management, actuarial thinking, data, AI, uncertainty, and decision-making. In Phase 4, you will turn these ideas into your own group-based Capstone Project.

The Capstone Project will lead towards the **Showcase Event scheduled for 17 October 2026**. If there is any change to the date, time, or venue, the programme team will announce the updated arrangement through the course website and contact emails as soon as possible.

1 What is risk again?

Opening trigger

Before we start Phase 4, let us ask again:

What is risk again?

Risk is not only about insurance, investment, accidents, or probability. Today, risk is also very close to you.

- You may not be able to find suitable groupmates.
- You may forget some of the knowledge learned in Phases 1 to 3.
- Your groupmates may become inactive, disappear, or communicate poorly.
- Your project discussion may clash with your normal school timetable.
- Your group may choose a topic that sounds interesting but is too broad.
- Your group may not find suitable real data.
- Your final presentation may not show clearly what you have learned.

Learning Moment 1: Risk is uncertainty with consequences

In Phase 4, your group project is also a risk-management problem. You face uncertainty about groupmates, topic selection, time management, communication, data, simulation, and presentation quality. Your job is not to avoid all uncertainty. Your job is to identify, assess, and manage it.

2 Your Phase 1 to 3 toolbox

You have already learned many tools. A good Capstone Project does not need to use all of them, but it should use suitable tools well.

- Probability
- Expected value
- LLM and CAT
- Financial instruments
- Risk measures
- Portfolio risk
- Diversification and hedging
- Insurance and annuity
- Actuarial pricing
- Statistical modelling using R
- Data visualisation
- Decision-making under uncertainty

Write one tool you may reuse

One concept, tool, or skill from Phases 1 to 3 that may help my project:

3 Forward and option in Phase 4

In finance, we have learned about **forward contracts** and **options**. Can these ideas help us understand Phase 4?

Concept	Finance meaning	Phase 4 meaning
Forward contract	An obligation to do something later under agreed terms.	Every group has an obligation to deliver a well-prepared presentation on the designated date of the Showcase Event in Phase 5.
Option	A right, but not necessarily an obligation, to choose.	Today, students have the right to find suitable groupmates and interested topics.
Risk management	Managing uncertainty and consequences.	Managing grouping, topic, timeline, communication, data, simulation, and presentation risks.

4 What is a good Capstone Project?

Learning Moment 2: A good project starts from a good question

A good Capstone Project does not start from a beautiful slide. It starts from a clear question.

A strong project should have:

- a clear and interesting question;
- a feasible scope;
- real data;
- simulation supported by actual data where possible;
- clear assumptions;
- evidence, charts, tables, or calculations;

- a risk-management or actuarial interpretation;
- a final message that the audience can remember.

Mini example: Is the highest return always the best?

Suppose Fund A has a higher average return than Fund B. Is Fund A automatically better? Not necessarily. A higher return may come with much higher risk. A good fund analysis should consider both return and uncertainty. This is why we need risk measures, scenarios, simulation, and clear interpretation.

5 Choose one project direction

Your group should choose **one** of the following two directions.

5.1 Option 1: Risk analysis of existing investment fund(s)

Your group may analyse one existing investment fund. If useful, you may also analyse more than one fund and compare their risk, return, volatility, or other features.

Possible questions:

- How risky is this fund?
- Is the return attractive after considering the risk?
- How does this fund compare with another fund?
- What type of investor may find this fund suitable?
- What may happen to the fund under different simulated scenarios?

5.2 Option 2: Invent an innovative insurance or annuity product

Your group may design a new insurance or annuity product. The product should address a real risk, need, or uncertainty.

Possible questions:

- What risk does this product manage?
- Who may need this product?
- What benefit does this product provide?
- How can data support the product design?
- How can simulation show possible outcomes for customers or the insurer?
- Why is the product innovative or meaningful?

Our first project direction

Our group is leaning towards:

- ☐ Option 1: Risk analysis of existing investment fund(s)
- ☐ Option 2: Innovative insurance or annuity product

Our possible project question is:

6 Real data, simulation, and responsible AI use

6.1 Real data is required

Your project must use real data. If your group conducts simulation, you still need to use actual data to estimate or justify the parameters.

Examples include fund price data, return data, benchmark data, market data, demographic statistics, mortality-related statistics, inflation data, interest-rate data, public statistics, or small survey data.

6.2 Simulation is required

Every group must include simulation. The simulation should help the audience understand uncertainty, possible outcomes, risk, return, claims, product performance, or future scenarios.

You must be able to explain:

- what you simulated;
- why simulation is useful;
- what assumptions you made;
- how actual data helped you choose parameters;
- what the simulation results show;
- how the simulation supports your conclusion.

Learning Moment 3: A model is useful only when assumptions are understood

A simulation or model is not automatically convincing. It becomes meaningful only when your group understands the assumptions, the data, and the limitations.

6.3 Responsible AI use

You are encouraged to use AI tools such as Google Gemini Pro to generate ideas, explore datasets, draft code, improve summaries, design visuals, or organise slides.

AI can assist, but your group must understand, verify, and cite your work

You may use AI to brainstorm topics, search for public datasets, suggest methods, draft or debug code, improve slide structure, refine visuals, and practise explanations.

You must understand the AI-generated output, verify facts and data sources, check calculations and code, cite or acknowledge important AI-assisted work where appropriate, and avoid direct copy-and-paste.

7 What should your group produce?

Output	Required?	Reminder
Presentation slides	Yes	Use Microsoft PowerPoint, Google Slides, Canva, or other suitable tools. AI can help organise and typeset, but clear thinking is essential.
Written report	No	Not required because the Showcase Event is interactive. Good documentation is still needed internally.
Poster	Optional	Useful if your group wants to introduce the project visually.
Video	Optional	Useful for attracting attention or showing results more vividly. It should support, not replace, your live explanation.
Data analysis / simulation	Yes	Use R, Excel, Google Sheets, Python, Google Colab, or other suitable tools. Explain what and why, not only code or formulas.
Member participation	Yes	Every member must be involved in the presentation. Division of work is not an excuse.

8 Forming your project group

Each project group should normally have **five students**. Students choose their own groups first. If a group has fewer than five members, the programme team may help coordinate afterwards and fit in remaining students who do not yet have a group.

If you are absent today

If you cannot attend the 12 August face-to-face session, you should actively communicate with classmates and identify suitable groupmates as soon as possible. If classmates have included you in a group, they should inform the programme team during the session.

If you do not attend the session and fail to form a group, the programme team will handle your case afterwards. However, you may risk being assigned to a group with an irregular group size.

8.1 Suggested temporary roles

Roles are preferred but not mandatory. They are only a starting point and can be adjusted later.

Role	Possible responsibility
Group coordinator	Keeps the group organised and reminds members of deadlines.
Data lead	Thinks about data sources, datasets, simulation, spreadsheets, or coding work.
Actuarial/risk analysis lead	Connects the project to risk management, insurance, annuity, actuarial thinking, or uncertainty.
AI/tools lead	Explores tools such as R, Excel, Google Sheets, Python, Google Colab, Canva, or AI tools.
Presentation lead	Plans the presentation flow and speaking arrangement.
Slides writer	Helps organise slides, visuals, and layout.

Roles are not walls

A role helps your group start working, but it should not become a wall between members. Every member should understand the project question, data, simulation, findings, and presentation message.

9 Project proposal template

Submission requirement

Each group must submit a written proposal to **Mr Michael Kan, Learning Support Specialist**, no later than **24 August 2026, Monday, 23:59**. Mr Michael Kan will comment on the proposal and discuss with each group during the first Zoom meeting on **26 August 2026, Wednesday, 13:30-14:45**.

1. Group information

Group number
Group contact person
Contact method

2. Group members and duties

No.	Student name	Main duty	Presentation part
1			
2			
3			
4			
5			

3. Project direction and title

- ☐ Option 1: Risk analysis of existing investment fund(s)
- ☐ Option 2: Innovative insurance or annuity product

Our proposed topic title is:

4. Motivation

Why is this topic interesting, important, or meaningful? Why should the audience care?

5. Real data plan

What real data, public dataset, fund record, market data, public statistics, or survey data may help your project?

6. Simulation plan

What will your group simulate? How will the simulation support your analysis?

7. Responsible AI use plan

How may your group use AI tools, and how will your group verify the output?

8. Timeline

Date / period	Planned work
12 Aug 2026	Form group and discuss initial topic.
24 Aug 2026, 23:59	Submit written proposal to Mr Michael Kan.
26 Aug 2026, 13:30-14:45	Discuss proposal with Mr Michael Kan in the first Zoom meeting.
After proposal feedback	Revise topic, data plan, simulation plan, and presentation direction.
Before Showcase Event	Prepare slides, analysis, optional poster/video, and rehearsal.
17 Oct 2026	Showcase Event, subject to updates if any.

10 How to make your project presentation effective

Detailed presentation training will continue in later Zoom sessions. For now, remember Keith's short framework, adapted for the Capstone Project.

Phrase	Capstone meaning
To motivate me	Explain why the topic matters.
To surprise me	Show an interesting insight, comparison, invention, or unexpected finding.
To impress me	Support your work with data, simulation, evidence, and clear reasoning.
To present in multiple ways	Use slides, charts, tables, examples, formulas, analogies, optional poster/video, or demonstration.
To summarize	End with a clear takeaway that the audience can remember.

Learning Moment 4: Good presentation turns analysis into a story

Your audience does not want to read your code line by line. Your audience wants to understand your question, evidence, insight, implication, and takeaway.

Optional presentation storyboard

Part	Your group idea
Opening hook	
Problem statement	
Why it matters	
Data or evidence	
Analysis	
Main finding	
Risk-management implication	
Final message	

11 Important dates and expectations

Date	Event / requirement
12 Aug 2026, Wednesday	Phase 4 face-to-face kick-off, review, project briefing, group formation, and topic discussion.
24 Aug 2026, Monday, 23:59	Written project proposal due to Mr Michael Kan.
26 Aug 2026, Wednesday, 13:30-14:45	First Zoom meeting for proposal comments and group discussion.
17 Oct 2026, Saturday	Showcase Event, subject to updated announcement if any.

Students who complete the Capstone Project, attend the Showcase Event, and achieve overall 80 percent attendance will be awarded the programme certificate. The top three teams will receive a trophy for each member.

Students who are unavailable for a session should submit a formal leave application as soon as possible. Delayed group formation may not be beneficial to project development because groups are expected to begin planning immediately after this session. For further information and updates, please refer to the course website.

12 Before you leave today

Checklist

- ☐ I know my group number.
- ☐ I know my groupmates.
- ☐ We have one possible topic.
- ☐ We know what data or examples we may need.
- ☐ We have assigned immediate next steps.
- ☐ We know what to do before the next session.

End note for programme documentation

This handout is prepared for OSALP Programme 2025-04 under the Gifted Education Fund: Off-school Advanced Learning Programmes. The Capstone Project forms part of the programme learning and showcase deliverables. Students should keep clear records of their project topic, data sources, assumptions, tools used, analysis steps, AI assistance, division of work, and presentation materials.